Multi-Source Chest X-Ray Dataset Preparation & Analysis

Integrating CheXpert and NIH ChestX-ray14 for Thoracic Abnormality Modeling

TOTAL DATASET SCOPE

303,349

Frontal-view radiographs

CHEXPERT CONTRIBUTION

191,229

Images

NIH CHESTX-RAY14

112,120

Images

Six Core Abnormality Labels Were Harmonized Across Sources

- **Harmonized Target Labels:** Pleural Effusion, Cardiomegaly, Pneumothorax, Atelectasis, Consolidation, and Lung Lesion.
- **Label Mapping:** The "Lung Lesion" target approximates the union of NIH's "Mass" and "Nodule" labels.
- **Flag Variable:** A seventh "No Finding" column serves strictly as a flag to separate clean normals, not as a training target.
- **Missing Data:** CheXpert blank values were imputed to 0. Uncertain labels (-1) were preserved in the manifest for dynamic load-time resolution.
- **NIH Constraints:** NIH labels remain strictly binary, where 0 indicates "not labelled positive" rather than "confirmed absent".

A Deterministic Pipeline Standardizes Inputs to a 224×224 Equalized Format

- **Grayscale Conversion:** All images are loaded and converted to single-channel grayscale.
- **Resizing Strategy:** Lanczos interpolation resizes images to 224×224 pixels. Aspect ratio is stretched rather than padded to prevent layout-based source identification.
- **Intensity Equalization:** Global histogram equalization is applied identically to all resized images.
- **Pretrained Normalization:** Inputs are scaled to [0, 1] using ImageNet mean and standard deviation.
- **Spatial Augmentation:** Training utilizes random affine transformations (rotation up to 7 degrees, translation up to 5 percent, scale 0.93 to 1.07). Horizontal flips are deliberately excluded to preserve left/right anatomical meaning.

Initial EDA Revealed Critical Differences in Intensity and Label Prevalence

Image Intensity Profiles

CheXpert (Equalized): Flat histogram (mean 129.15, SD 73.75) indicates global equalization prior to release.

NIH (Natural): Demonstrates natural intensity variation (mean 122.50, SD 60.91).

Pleural Effusion Prevalence

40.25% vs 11.88%

CheXpert exhibits over 3x higher prevalence of Pleural Effusion compared to the NIH dataset.

Cardiomegaly Variance

12.26% vs 2.48%

Cardiomegaly shows a 12.26% prevalence in CheXpert, compared to just 2.48% in NIH.

Label Distribution

Negative Class Dominance

In CheXpert, blank values heavily dominate the negative class, requiring carefully designed imputation strategies during preprocessing.

Strict Patient-Level Partitioning Ensures Zero Data Leakage

- **Partitioning Strategy:** Splits are grouped strictly by patient ID, guaranteeing no patient appears in multiple partitions.
- **Training Split Methodology:** The official training lists for both NIH and CheXpert were split 90/10 into training and validation sets using a fixed random seed.
- **CheXpert Training Volume:** 171,927 images distributed across 58,080 distinct patients.
- **NIH Training Volume:** 77,988 images distributed across 25,207 distinct patients.
- **Dedicated Test Sets:** The official NIH test list (25,596 images) and the CheXpert official validation set (202 images) were retained entirely as dedicated test splits.

Datasets Remain Highly Distinguishable via Residual Image Statistics

0.990

ROC AUC • RESIDUAL LEAKAGE

Models can still separate sources using combined intensity and sharpness features.

0.906

ROC AUC • PADDING MITIGATION

Stretching images successfully reduced the layout identification cue down from an ROC AUC of 0.974.

- **Separability Testing:** A gradient-boosted classifier was trained to predict the image source using 5-fold cross-validation.
- **Sharpness Discrepancy:** CheXpert retains a residual sharpness approximately two times higher than NIH due to differing downsampling reduction factors.
- **Performance Risk:** Source predictability may artificially inflate pooled model performance due to differing baseline label prevalences.

CheXphoto: Evaluating Mobile Radiograph Capture

- **Dataset Purpose:** CheXphoto extends CheXpert with smartphone photos and digital synthetic transformations to benchmark real-world mobile capture.
- **Capture Variability:** Evaluates model degradation under smartphone artifacts including ambient glare, perspective tilt, lens blur, and varied lighting.
- **Validation Benchmark:** Includes radiologist-adjudicated labels across 663 test photos, measuring the domain gap between direct PACS scans and mobile captures.

CLINICAL DEPLOYMENT IMPLICATION

Essential for point-of-care tele-radiology where diagnostic models must remain robust when clinicians photograph physical films or monitors using consumer mobile devices.